

Zero-trust security for brownfield OT/ICS sites in energy utilities.

Energy × OT and industrial control system security × Zero trust architecture · OS038 v2 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
71 is the world moving	71 can we play, can we win	MEDIUM 4 named in the evidence	€286m observed evidence · per year	NEXT regulation adopted or propose...	L0 15 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

Evidence gap (SC-13). Orange has 3.0 published reference(s) in Energy, below the threshold of 5. A customer conversation here has no proof point behind it yet.

The opportunity

This opportunity is to deliver zero-trust security for brownfield OT/ICS sites in energy utilities, addressing persistent gaps exposed by recent cyberattacks on critical infrastructure. Energy utilities with legacy industrial control systems need a pragmatic, staged approach to zero trust that respects their existing equipment and regulatory pressure. The time is now because regulators are moving toward binding directives and operators are prioritizing visibility, segmentation, and operational resilience.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€347m €77.5m – €4.4bn	€286m €63.9m – €3.6bn	€17.1m €3.8m – €217m	observed
Observed: tendered contract value, annualised	€172m €172m – €460m	€172m €172m – €460m	€10.3m €10.3m – €27.6m	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Energy (EU27_2020)	5,796 enterprises	observed	Eurostat · sbs_sc_ovw	2024
Enterprises adopting Zero trust architecture	86.3 % of enterprises (10+ employees)	observed	Eurostat · isoc_cisce_ran2 · E_SECMNAC	2024
Annual value of one engagement	€299k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-07
Size-weighted buyer base	1,341 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	6.0 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders.

Observed: tendered contract value, annualised

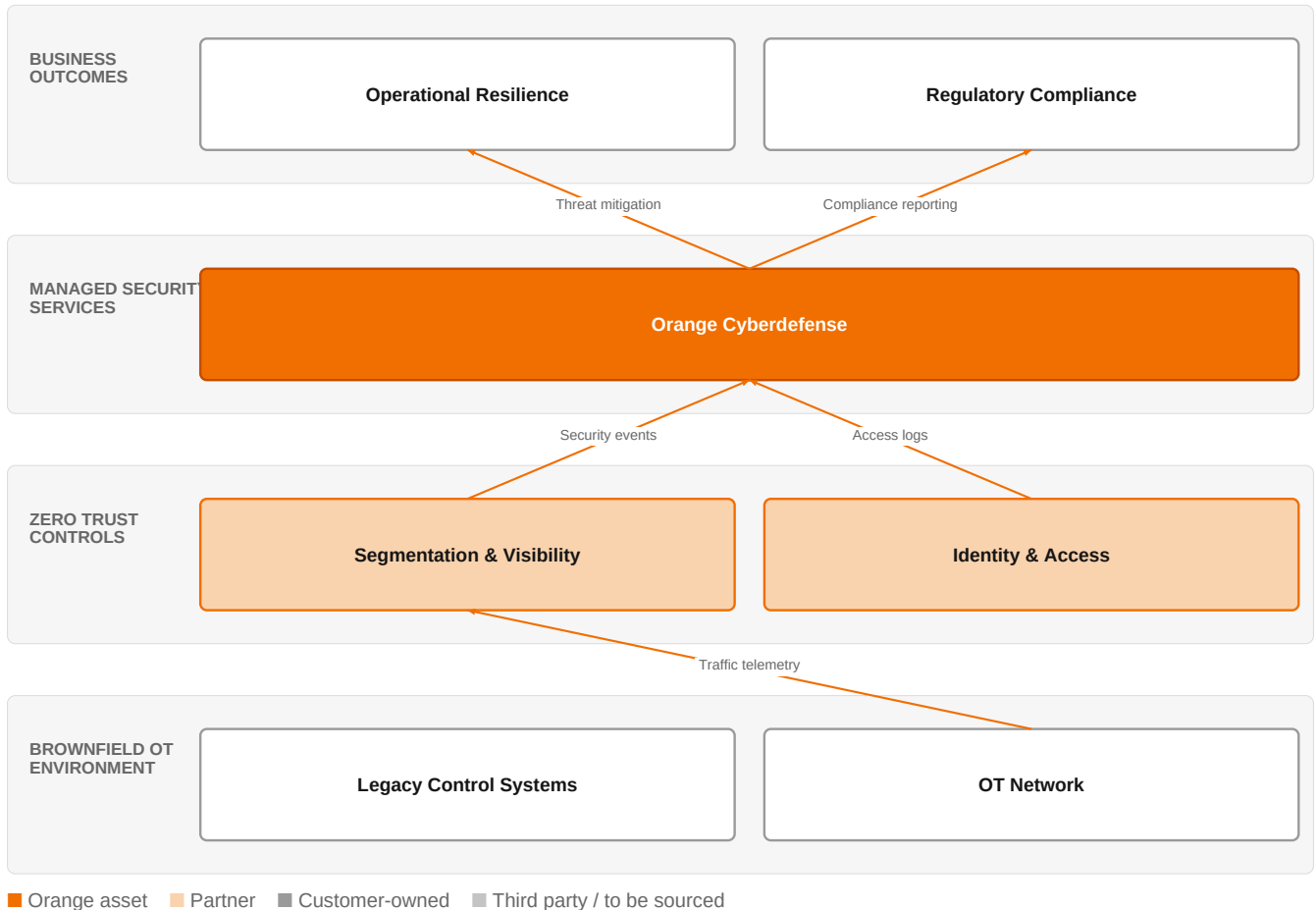
Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€172m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-07
Assumed obtainable share of the serviceable market	6.0 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 21 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a

market size.

The solution, in one picture

Zero Trust for Brownfield OT



This picture shows how Orange orchestrates zero trust for brownfield OT without replacing legacy systems, using managed services and partner controls to achieve resilience and compliance.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

Recent cyberattacks on the water sector have exposed persistent OT security gaps, prompting the OTCC to urge CISA to issue a binding OT cybersecurity directive. Brownfield sites have a unique mix of legacy equipment and control systems that cannot simply be replaced, making complete transformation daunting and risky. Zero trust in OT is moving beyond identity as industrial operators prioritize visibility, segmentation, and operational resilience. Regulators including NIS2, DORA, CISA, and SAMA are converging on zero trust as the security standard.

Sources: SIG-49028056B8B3 GovCIO Media & Research 2026-08-12; SIG-FCFE20A18D9A Industrial Cyber 2026-08-03; SIG-E2034B250974 bing.com 2026-08-13; SIG-CFBD245A217D Industrial Cyber 2026-05-24; SIG-B3D9AD374F05 Atos 2026-05-06

Who buys, and why now

The chief information security officer (CISO) or OT security manager signs, but the plant operations manager feels the pain of unplanned downtime and safety risks from insecure legacy systems. The trigger is often a near-miss or actual cyber incident, an audit finding, or a regulatory directive from bodies like CISA. They need to improve security without disrupting continuous operations or replacing expensive, working equipment.

Sources: SIG-49028056B8B3 GovCIO Media & Research 2026-08-12; SIG-FCFE20A18D9A Industrial Cyber 2026-08-03; SIG-E2034B250974 bing.com 2026-08-13

Why it is hot now — every claim bound to a dated source

- Recent cyberattacks on the water sector expose persistent OT security gaps. [SIG-49028056B8B3]

- The OTCC urges CISA to issue a binding OT cybersecurity directive after cyberattacks disrupted water utilities. [SIG-FCFE20A18D9A]
- Cyberattacks on the water sector expose persistent OT security gaps, indicating a pressing need for improved OT security. [SIG-49028056B8B3]
- Brownfield sites have a unique mix of legacy equipment and control systems that cannot simply be replaced, making zero-trust approaches necessary. [SIG-E2034B250974]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a team of cybersecurity experts and defense and security experts to conduct a brownfield OT assessment, leveraging Orange Group researchers for threat intelligence. The engagement would start with a discovery phase to map the legacy OT environment, then design a zero-trust architecture that uses segmentation and visibility, and finally deploy Orange Cyberdefense managed services for continuous monitoring and response. The solution would be delivered on ISO 27001 and SecNumCloud certified infrastructure, with optional integration of partner technologies from Palo Alto Networks, Fortinet, or Zscaler.

Why Orange can win

Orange can win because it combines deep cybersecurity expertise with a sovereign, certified delivery model (ISO 27001, SecNumCloud) that is critical for energy utilities. The ability to orchestrate a pragmatic, staged approach for brownfield sites, rather than a rip-and-replace, is a differentiator. However, the assets are thin on specific OT domain expertise, so Orange must leverage its partner ecosystem to fill gaps.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Orange Cyberdefense managed services	offer	L0	offer addresses the use case AND provides the technology	unconfirmed
Palo Alto Networks	partner	L2	partner provides the required technology	unconfirmed
Fortinet	partner	L2	partner provides the required technology	unconfirmed
Zscaler	partner	L2	partner provides the required technology	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
SecNumCloud	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
Cybersecurity experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Defense and security experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed

Competitive landscape

MEDIUM competitive intensity (score 7.4; 4 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Palo Alto Networks and Zscaler are strong in zero-trust architecture and are also Orange partners, so they may be both competitors and collaborators. Microsoft brings hyperscale cloud and identity capabilities, while Atos/Eviden is a global systems integrator with OT experience. CrowdStrike, Fortinet, Cisco, and Bechtle also sell zero-trust solutions, with Fortinet and Cisco strong in networking and security. The customer will compare Orange's managed service and sovereign credentials against these specialists.

Sources: SIG-10E1BC61FB2F Palo Alto Networks 2026-05-12; SIG-46B6362BF559 IOT Insider 2026-04-30; SIG-333EF44491DD Microsoft 2026-04-29; SIG-B3D9AD374F05 Atos 2026-05-06

Competitor	Category	Position here	Basis	Evidence
Palo Alto Networks	Security specialist	sells Zero trust architecture; competes in Cybersecurity — also an Orange partner	evidenced	SIG-10E1BC61FB2F
Zscaler	Security specialist	sells Zero trust architecture; competes in Cybersecurity — also an Orange partner	evidenced	SIG-46B6362BF559

Microsoft	Hyperscaler	named in this topic's evidence — also an Orange partner	evidenced	SIG-333EF44491D D
Atos / Eviden	Global systems integrator	competes in Cybersecurity	evidenced	SIG-B3D9AD374F 05
CrowdStrike	Security specialist	sells Zero trust architecture; competes in Cybersecurity	structural	—
Fortinet	Security specialist	sells Zero trust architecture; competes in Cybersecurity — also an Orange partner	structural	—
Cisco	Network equipment vendor	sells Zero trust architecture; competes in Cybersecurity — also an Orange partner	structural	—
Bechtle	Regional integrator or reseller	sells Zero trust architecture	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 Have you experienced a recent cyber incident or audit finding that highlighted OT security gaps?
- 2 What is the age and mix of your legacy control systems, and are any of them end-of-life?
- 3 Do you have a current inventory of all OT devices and network segments?
- 4 What regulatory frameworks (e.g., NIS2, CISA directives) are you currently preparing for?
- 5 How do you currently handle remote access to OT systems, and is it VPN-based?
- 6 What is your tolerance for downtime during security improvements?

Objections you will meet

They will say	You can answer
We can't afford to shut down production to implement zero trust.	That's a valid concern. Our approach is designed for brownfield sites, using phased segmentation and visibility that minimizes disruption. We start with monitoring and then incrementally enforce policies, so you don't have to stop operations.
Our legacy systems can't support modern security controls.	True, many legacy systems lack the ability to run agents. That's why we use network-level segmentation and visibility, and we can deploy non-intrusive sensors. We work with what you have, not what we wish you had.
We already have a security vendor; why do we need Orange?	You may have point solutions, but zero trust for OT requires a holistic, managed approach. Orange brings together expertise, certified infrastructure, and a partner ecosystem to orchestrate a solution that fits your brownfield reality, not just a product.

Risks and unknowns

The main risk is that brownfield OT environments are highly heterogeneous and complex, making a one-size-fits-all zero-trust model difficult to apply. There is also uncertainty about the timeline and specifics of binding regulatory directives, which could change priorities. Orange's lack of deep OT-specific expertise may be exposed if partners are not effectively integrated.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Develop a bid brief for a European energy utility that outlines a zero-trust architecture for brownfield OT/ICS, incorporating Orange Cyberdefense managed services, and highlight our ISO 27001 and SecNumCloud certifications.
Sales	In a conversation with a European energy utility's CISO, reference Orange Cyberdefense managed services and our partnerships with Palo Alto Networks, Fortinet, and Zscaler to discuss how we can secure their legacy OT environments.
Strategist / Innovator	Commission a deep-dive brief on zero-trust reference architectures for brownfield OT/ICS, leveraging Orange Group researchers and cybersecurity experts, and map them against ISO 27001 and SecNumCloud compliance requirements.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-0CB746316F44	2026-05-21	International Journal of ...	1	Industrial Control System Cyber Security Frameworks: Assessment using Defense-in-Depth Criteria
SIG-2CC53D62856D	2025-08-18	openalex.org	1	A Hybrid Zero Trust Deployment Model for Securing O-RAN Architecture in 6G Networks
SIG-E2034B250974	2026-08-13	bing.com	2	SolutionsPT: Inside the new industrial cyber security playbook
SIG-49028056B8B3	2026-08-12	GovCIO Media & Research	2	Cyberattacks on Water Sector Expose Persistent OT Security Gaps - GovCIO Media & Research
SIG-FCFE20A18D9A	2026-08-03	Industrial Cyber	2	OTCC urges CISA to issue binding OT cybersecurity directive after cyberattacks disrupt water ut...
SIG-74A5048912D7	2026-07-10	Robotics & Automation News	2	How Zero Trust Network Access Eliminates Implicit Trust at Scale - Robotics & Automation News
SIG-5A3260810D62	2026-06-08	FedTech Magazine	2	Machine Identity Management: The Nonhuman Side of Federal Zero Trust - FedTech Magazine
SIG-CFBD245A217D	2026-05-24	Industrial Cyber	2	Zero trust in OT moves beyond identity as industrial operators prioritize visibility, segmentat...
SIG-D2DC2DA0537D	2026-05-14	appinventiv.com	2	Zero Trust Architecture Implementation in Australia - appinventiv.com
SIG-10E1BC61FB2F	2026-05-12	Palo Alto Networks	2	Healthfirst Applies an Identity Security-First Approach to Implement Zero Trust - Palo Alto Net...
SIG-7E74F5F920F4	2026-05-12	EY	2	OT Control Tower - Zero Trust for Industrial Environments - EY
SIG-B3D9AD374F05	2026-05-06	Atos	2	NIS2, DORA, CISA, SAMA: Why Zero Trust Became the Security Standard Regulators Agree On - Atos
SIG-08B271ABAD4A	2026-04-30	iTnews	2	State of Security 2026: Zero Trust - iTnews
SIG-46B6362BF559	2026-04-30	IOT Insider	2	IXT integrates Zscaler Zero Trust into cellular IoT to replace VPNs and strengthen OT security ...
SIG-333EF44491DD	2026-04-29	Microsoft	2	Scale AI safely with Zero Trust security - Microsoft

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:13+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:17+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-18T20:42:40+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.